SUBRAHMANYESWAR KOLLURU

AI / ML Engineer | Computer Vision & LLM Systems

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PROFESSIONAL SUMMARY

AI/ML engineering student experienced in building intelligent systems across computer vision, LLM applications, and AI automation. Experienced in developing real-time AI pipelines, multi-agent systems, and retrieval-augmented generation architectures. Published researcher at ICCCNT 2025 (IIT Indore) and Top 10 finalist in a national project expo. Strong interest in building scalable AI systems involving model optimization, edge inference, and full-stack AI deployment.

TECHNICAL SKILLS

AI / ML	Python, TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, YOLO, ArcFace, MobileNetV2, LSTM, Transfer Learning
LLM & GenAI	LangChain, LlamaIndex, Ollama, Llama 3, OpenAI Whisper, RAG Pipelines, Vector Databases, Embeddings, LoRA / PEFT, HuggingFace, Fine-Tuning
AI Agents	CrewAI, Multi-Agent Systems, MCP Protocol, Agent Orchestration, Tool Integration
Deployment	FastAPI, TensorRT, Intel OpenVINO, FP16 Quantization, Edge AI, Docker (basic), Streamlit, Next.js, Redis, WebSocket
Data & DB	SQL, MongoDB, Neo4j, NumPy, Pandas, Matplotlib
Dev Tools	VS Code, Jupyter Notebook, Git, GitHub

WORK EXPERIENCE

AI Engineering Intern | READ Automation Pvt. Ltd., Chennai — *Dec 2025 – Jan 2026*

Project: github.com/Subrahmanyeswar/Bottle-Cap-Inspection-System

- Developed an automated optical inspection (AOI) pipeline using Python, OpenCV, and TensorFlow for real-time manufacturing defect detection.
- Applied transfer learning on MobileNetV2 achieving 95% classification accuracy with 100% recall on critical safety defects.
- Optimized inference with Intel OpenVINO & FP16 quantization, boosting throughput by 23.8% to 447 FPS at 2.23ms latency.
- Implemented motion-based triggering via frame differencing, cutting idle CPU overhead by 60%; applied CLAHE preprocessing for robust detection across 200–2000 lux lighting.

PROJECTS

Aeternum Hub — Real-Time AI Security System | *Python, YOLO, TensorRT, ArcFace, Moondream2, FastAPI, Redis, Next.js*

GitHub: github.com/Subrahmanyeswar/aeternum-hub

- Architected a full-stack AI security platform with <100ms detection latency, 95%+ face recognition accuracy, and 3–5x TensorRT GPU speedup.
- Integrated YOLO + ArcFace for real-time multi-person tracking and threat classification across 8 object classes; implemented multi-stage progressive alert system (5s/10s/20s).
- Built Moondream2 vision-language pipeline for automated AI-generated police reports; designed a privacy-first architecture with fully local AI inference and processing.

Multi-Agent Competitive Intelligence Platform | *CrewAI, Python, LLMs, Multi-Agent Systems*

GitHub: github.com/Subrahmanyeswar/multi-agent-competitive-intelligence

- Designed a multi-agent orchestration pipeline with specialized Research, Analysis, and Synthesizer agents for autonomous competitive landscape monitoring.
- Implemented parallel information processing with structured weekly intelligence report generation using agent-to-agent task delegation.

Advanced RAG System | *Vector DB, Llama 3, Ollama, Qdrant (Hybrid Search), FlashRank (Cross-Encoder), LangChain*

GitHub: github.com/Subrahmanyeswar/Aeternum-AI-Advanced-RAG

- Engineered an end-to-end pipeline using Qdrant to merge 1024-dimension dense embeddings with sparse keyword vectors, ensuring the system captures both deep semantic meaning and exact technical terminology.
- Implemented a high-performance 3-stage workflow consisting of Stage 1 Recall (40 candidates), Stage 2 RRF Fusion (merging to 15 chunks), and Stage 3 FlashRank Reranking to deliver highly accurate, grounded responses from local **Llama 3** inference.

Jarvis AI Assistant | *Python, Llama 3, OpenAI Whisper, YOLO, ONNX*

GitHub: github.com/Subrahmanyeswar/Jarvis-AI-Assistant

- Developed a voice-controlled AI assistant integrating Llama 3 for NLP, Whisper for speech recognition, and YOLO-based face recognition for intruder detection and security alerts.

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) — *CGPA: 8.27 | 2023 – 2027*

Kalasalingam Academy of Research and Education (KARE)

ACHIEVEMENTS

- Research Publication — ICCCNT 2025, IIT Indore (AI Security)
- Hackathon — 3rd Prize, national-level AI/ML competition

- National Project Expo — Top 10 finalist among 100+ teams